

Raihan Ahmad

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Professional Summary

Software engineering student skilled in MERN and FastAPI, with a strong foundation in system design. Experienced in building REST APIs, managing PostgreSQL, and deploying containerized services using Docker and Kubernetes.

Technical Skills

- **Programming Languages:** Python, JavaScript, Java, SQL, C
- **Backend:** FastAPI, Node.js, Express.js
- **Frontend:** React.js, Tailwind CSS, Bootstrap
- **Databases:** PostgreSQL, MySQL, MongoDB, Firebase Firestore
- **DevOps & System Design:** Docker, Git/GitHub, AWS EC2, REST APIs, OAuth2/JWT, Postman

Experience

Software Developer Intern

Mar 2026 - Present

Nucelux Private Limited

- Built and scaled a production-grade **HRMS & Finance platform** using FastAPI, delivering 300+ async REST APIs.
- Designed 70+ optimized PostgreSQL tables for payroll, attendance, and approval workflows.
- Engineered a role-based approval system with multi-level workflows, audit trails, and JWT-based RBAC.

Full-Stack Developer Intern

Mar 2025 - Apr 2025

Zidio Development

- Led front-end development of **FinTrak** , building intuitive React.js and Material UI components.
- Assisted in developing secure REST APIs using Spring Boot and MySQL with JWT-based authentication.
- Integrated analytics dashboards using Chart.js to improve usability and insights.

Projects

Delivery Route Optimizer (CVRPTW)

Mar 2026 - Apr 2026

- Developed a CVRPTW route optimization system using React and FastAPI, leveraging ACO, GA, and PSO to reduce route distance by 28–43% and constraint violations by up to 91%.
- Added benchmarking, constraint handling, and interactive visualization using Leaflet and Chart.js.

AI-Powered Student Stress Detector

Nov 2025 - Dec 2025

- Engineered a stress analysis system using React + FastAPI with a Mamdani FIS (21 rules) for real-time inference.
- Implemented offline fallback, persistent history, and Chart.js visualizations for low-latency analysis.

UniBazaar - Campus Marketplace

Jan 2025 - Feb 2025

- Developed a marketplace (React, Node.js, MongoDB) supporting authenticated listings and CRUD operations.
- Integrated Firebase Auth, Cloudinary storage, and deployed services via Netlify and Render.

Education

B.Tech in Computer Science and Engineering

2023 - 2027

Birla Institute of Technology, Mesra CGPA: 8.43

Senior Secondary (Class XII), CBSE

2022

Holy Cross School, Hazaribagh 91%

Matriculation (Class X), CBSE

2020

Holy Cross School, Hazaribagh 95.5%

Leadership & Activities

Semifinalist, Trikaya Hack-to-Hire Hackathon

Nov 2025

Recognized for developing **PodcastMini** , an AI-powered podcast summarization tool.

Google Student Ambassador

Aug 2025 - Dec 2025

Conducted AI workshops and engaged students in innovation initiatives.

General Secretary, Aakshar Literary Society

Jan 2025 - Nov 2025

Led major campus events and contributed to growing participation and outreach.

School Captain, Holy Cross School

2020 - 2022

Directed the student council and coordinated school-wide events.